

Integrating IoT and Machine Learning for Mobile and Wearable Heat Risk Tracking in Outdoor Sports

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Background: Hyperthermia, including heat stroke, is a critical threat for millions of athletes in summer. It is a leading cause of preventable deaths and ER visits in college and high school sports. To assess heat risk and adjust athletic activities, international, national and state athletic associations have adopted the Wet Bulb Globe Temperature (WBGT) in their safety policies. WBGT quantifies the heat stress in direct sunlight by measuring solar radiation, temperature, humidity, and wind speed. These heat safety policies, unfortunately, are not used often in schools because WBGT meters are expensive (>\$500ea). Thus, several mathematical models have been developed to estimate WBGT with meteorological parameters. However, some of the parameters, such as solar radiation, are not commonly available for athletes and coaches.

Objective: This work proposes a method to estimate a heat alert level from a set of meteorological parameters readily available for athletes and coaches. It employs machine learning (ML) in compliance with a major heat safety policy. This work also implements the proposed ML method in mobile and wearable IoT devices to continuously track the current alert level during an athletic activity and issue early warnings for precautions and activity adjustment.

Related work: Existing ML-based work for heat risk estimation predict WBGT from solar radiation data, periodic time-series data, and personal, real-time physiological data such as heart rate. The proposed method is different from them in that it does not rely on solar radiation, time series, nor personal physiological data.

Methodology: This work uses the decision tree (DT) and random forest (RF) algorithms to build supervised classifiers, which take a set of meteorological parameters as an input: temperature (F), relative humidity (%), cloud cover (%), precipitation (inch), wind speed (knot), and time of day. The classification output is one of the heat alert levels that are defined by Grundstein et al. and adopted by the National Federation of State High School Associations. The proposed classifiers are integrated with battery-operated RP2040 and ESP32 microcontrollers to periodically download meteorological parameters from online weather services and update the current alert level through deep sleep cycles. The parameters and alert level are shown on a batteryless e-ink display. When the alert level is in the “high risk” or “extreme” condition, a 100dB audible alarm is raised with a speaker.

Results: The proposed classifiers are trained with 46,429 meteorological data samples in the summer of 2024, which were collected from 290 cities in the US with an online database of the National Oceanic and Atmospheric Administration (NOAA). After hyperparameter adjustment, the DT and RF classifiers yield 91% and 99% classification accuracy in five-fold stratified cross-validation. The proposed IoT devices have been empirically tested in tennis practice and matches. By saving their power consumption effectively, the devices can run over three hours, which is long enough for a practice and match. Their audible alarm is highly noticeable even in intensive practice and matches.

Keywords: Hyperthermia, heat safety, wet bulb globe temperature (WBGT), multiclass data classification, Internet of things